

to load the IPv6 module by executing 'modprobe ipv6.' If successful, this module should be listed by the following command:

```
# lsmod |grep -w 'ipv6' && echo "IPv6 module successfully loaded"
```

It is also possible to automatically load the IPv6 module on demand. You have to add the following line to the kernel module-loader configuration file, which is normally '/etc/modules.conf' or '/etc/conf.modules':

```
alias net-pf-10 ipv6 # automatically load IPv6 module on demand
```

If your kernel has no IPv6 support you can try one of the following options:

1. Update your Linux distribution to a current one that supports IPv6 out-of-the-box.
2. Compile a new vanilla kernel.
3. Recompile kernel sources given by your Linux distribution.
4. Compile a kernel with third-party IPv6 extensions.

However, it is pertinent to point out

that IPv6 support in the kernel and the other basic packages is pretty average in terms of performance, and does not conform to all drafts and request for comments (RFCs). Most people tend to switch off the IPv6-compliance feature, as it makes the system quite slow.

To overcome this, you can make use of the Universal Playground for IPv6 (USAGI) kit, which contains an enhanced IPv6-enabled version of both the latest 2.6.x Linux kernel, and the most important networking software packages for Linux. USAGI is a project run by volunteers from various organisations, and aims to develop a production-quality IPv6 protocol stack and IPsec for Linux. Based in Japan, USAGI teams up with other such projects like the WIDE Project, the KAME Project and TAHL, to implement all absent or obsolete IPv6 support in Linux. The latest release of the USAGI kit can be downloaded from <http://www.linux-ipv6.org/>.

Apache and IPv6

The Apache HTTP server supports IPv6 on most platforms, allowing Apache to allocate IPv6 sockets and handle requests sent over IPv6. However, to allow the IPv6 socket to handle both IPv4 and IPv6 connections, you will need to tweak the configuration parameter and use generic Listen directives. To verify that Apache is working in dual-stack mode (supporting both IPv4 and IPv6), use the netstat command.

The writing is on the wall. With IPv4 addresses having been exhausted, the time, cost and complexity of operating a stable and growing Internet will rise significantly, unless IPv6 deployment is well under way. So hurry and make your organisation IPv6-ready, now! ●

The author is an IT consultant based in Faridabad, Haryana. The article has been reproduced from EPY's Linux For You magazine

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